

Sanghamitra

సంఘమిత్ర

(విజ్ఞాన, వినోద, వికాస త్రైమాసిక పత్రిక)

(A quarterly magazine meant for knowledge, entertainment and progress)

<http://www.sanghamitra.org>

WEDNESDAY, JULY 1, 2009



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Foreword

Sanghamitra wishes good luck to all students to achieve better performance in their fields in this academic year. Sanghamitra is being released with few articles this time due to some other commitments. Next issue will be released as usual with variety of articles. Sanghamitra English version completed Two years.

After Mathematics Seminar and Medhavadhanam 2009 in St. Louis, many participants enjoyed the program and asked for Summer classes. Due to India trip, we conduct short session during Spring break and we resume it again in next year. We happened to conduct classes in bay area in different areas and different age groups. The response was amazing and the attendees left with a great satisfaction. We are conducting classes in Visakhapatnam also for a week for the second time.

Received mails regarding the cover page with Sita & Rama's picture, Kids' corner, Salutation to your solution etc. We again brought 'Mathematrix' article in this issue with a minor change 'Mathematrix' instead of Mathematrix. Hope you like it. Give us your feedback on this.

Whatever Sanghamitra is contributing so far is a sand particle at a sea shore. There are so many other scholars who can contribute a lot and make a difference. Here the persons sharing knowledge and receiving knowledge both should be in a good understanding. There should not be any clash. When this act is done with devotion and as a blessing, both parties attain better state. While sharing age does not come into picture. A Seven year old child may know more than a Forty year old person. Keeping the ego aside, if the person opens up, he can learn a lot. The attitude of a learner should be to open up and hear what the other person says. If you want to share your ideas, please send mail to mathematrix@gmail.com or sanghamitra@gmail.com.

Let us meet again in the next issue due in October.

-Sanghamitra Team.

Learning while having fun!



Participants of the Vedic Mathematics Seminar.

We never feel tired when we do something which is interesting. While watching movie, even if we feel sleepy, we keep watching - ignoring the sleep. A kid forgets about food and other activities if he finds some entertainment. No need to say in modern days video

games became life to many kids.

On May 8th, Friday Sanghamitra organized a seminar on Vedic Mathematics. Audience were spell bound by hearing the formulae. Some of the participants showed interest in learning those techniques online if Sanghamitra provides online demonstration.

When the problem solving techniques are given in fun way, kids not only learn them, but also like to explore new things. Basically these seminars gives confidence in them, not the solution for each and every problem. Once the foundation is strong, building super structure is easy. More than Eighty audience participated in that program. Later Sanghamitra shared the Vedic Mathematics knowledge as well as problem solving techniques in Four other different places namely San Jose, Sunnyvale, Cupertino, Santa Clara etc. The idea behind these sessions is to encourage the participants to share what they know with others, by which they acquire more knowledge.

Let us meet again in October issue. Till then good bye!

- SuSri.

Words of Wisdom

Catch someone doing something right

With the blessings of Didi Ji, publishing her simple and strong words through 'Words of Wisdom', which are capable of transforming a person and lead them towards enlightenment. I know 'Words are powerful', but I experienced the fact in Didi Ji's presence. Let's say WOW! & Imbibe the 'Words of Wisdom'. May her blessings enlighten many souls and change the world into heaven. Didi Ji (Shri Siddheshvari Devi) is the founder of Radha Madhav Society. Visit www.radhamadhavsociety.org for a listing of Didi Ji's upcoming lectures and retreats.

It is very important to give recognition for a job well done. Usually we criticize immediately and hold off on the praise, but we should do the opposite. When you catch your employee, your child, your student, your spouse or your subordinate doing something right, give a nod, a wink, a token of appreciation or a pat on the back to show that you have noticed them doing something right.



When a parent calls out to a child, the child asks, "What did I do now?" Imagine how wonderful it would be for the child to hear, "I just want to tell you that you are a really good kid; you are very helpful and respectful, and I really appreciate it." I once asked all the children present in a retreat to see me in my room. They all came in expecting a scolding for perhaps being too loud or not following the rules. But instead of getting a reprimand, they got some genuine words of appreciation and a treat for good behaviour.

To be most effective, recognition must be immediate. When a teacher tells the student, "I really liked the hard work you put in that essay four months ago," the student scratches his head to figure out which essay he is being complimented on. To motivate the recipient to the maximum, you have to praise right away, not days or weeks or months down the line. The recognition must be specific also. There is a difference between telling a child, "I like your drawing" and "I like the way you used the pink and the blue in your drawing. I also like the way you drew the sun with a happy face."

Recognition is a very effective morale-booster and leads to greater productivity. Everyone needs to be appreciated and recognized. So go ahead and catch someone doing something right. *

Mathematricks

- Sreenivasa Rao Ainapurapu.

In the previous issue we ended with a question, "There was a pond. One day there was one lotus, next day there were 2, third day there were 4. In 30 days the pond was full of lotus flowers. Can you tell (i) in how many days the pond was half filled with lotus flowers. (ii) In how many days the pond was one-fourth filled with lotus flowers?" Hope you found the answer. If not, the pond was half filled with lotus flowers in 29 days and the pond was one-fourth filled with lotus flowers in 28 days.

***Note :** You should teach these techniques to kids only after they are ready to understand these methods. Then only they can appreciate these tips and the concepts behind. It is always better to use basic concrete methods to verify till they are comfortable.*

Some of Sanghamitra volunteers in bay area, California organized **Srinivasa Ramanujan Mathematricks Club** seminars on May 8th, 16th (in 2 different locations) and 17th (3 different locations) in different places. The participants enjoyed those Vedic Mathematics formulae. Sanghamitra appreciates all the volunteers for their time in organizing the events. We are going to bring some of the topics covered in bay area.

Solving 3×3 matrix in such a way that the sum of numbers in any row, any column and any diagonal is equal to the given number. The number should be a multiple of 3 and greater than or equal to 15 (to get all positive integers).

Some of you already know the solution. How do you explain this to kids? If you take numbers from 1 through 9 to fill the nine places, then sum of all the digits is equal to $1 + 2 + \dots + 9 = 45$. You have 3 rows. Sum of the digits in all the three rows is equal to 45. So each row should have a sum 15. Similarly each column and diagonal should have a sum equal to 15.

Now if we arrange numbers in the way shown (Figure A) below. You can see 4 possible sets of (1, 9), (2, 8), (3, 7) and (4, 6), the sum of which comes to 10. Adding the remaining number 5 to each of those sets bring the sum to 15. So 5 should be at the center. For example, $1 + 9 + 5 = 15$. The four possible combinations are shown in Figure B below.

Figure A

$$\begin{array}{l} \textcircled{1} + \textcircled{9} = \textcircled{10} \\ \textcircled{2} + \textcircled{8} = \textcircled{10} \\ \textcircled{3} + \textcircled{7} = \textcircled{10} \\ \textcircled{4} + \textcircled{6} = \textcircled{10} \\ \textcircled{5} \end{array}$$

Figure B

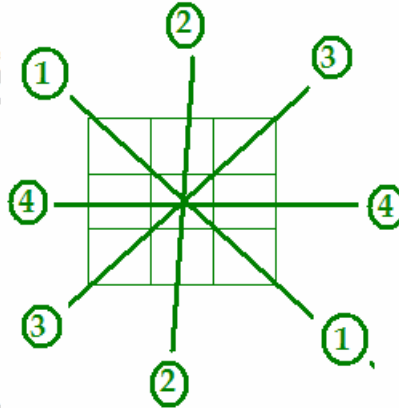


Figure C

8	1	6
3	5	7
4	9	2

If you see the solution in Figure C, the combination 1 is $(8+5+2)$, combination 2 is $(1+5+9)$, combination 3 is $(4+5+6)$ and combination 4 is $(3+5+7)$. The other combinations we need to look for are first column, third column, first row and third row; which are not connected to the middle 5. This kind of thinking will lead to better solutions in variety of problems.

As an extension to this we can take any number multiple of 3 and greater than or equal to 15. For example, let us take 27. So the center number should be 9. The rest of the numbers should be 5, 6, 7, 8, 10, 11, 12 and 13. That means Four numbers on either side of 9. So the rows are $\{12, 5, 10\}$, $\{7, 9, 11\}$ and $\{8, 13, 6\}$ respectively.

Isn't it interesting? Now you can try for other numbers, which are multiples of 3 and greater than or equal to 15. Verify your answers.

Find the value of $\left\{ 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots + \frac{1}{99} \right\}$

(Let us continue in the next issue, till then good bye!)

Smile if you wish

Thanks for curing!

Patient : Thank you doctor for curing me of gambling. You did a great job!

Psychiatrist: You are welcome! Here is my bill.

Patient : I will toss you for it – double or nothing!

Psychiatrist: ??? ☹

Cavities!

A dentist, golf player, was about leave his office with a golf bag on his shoulder. He got a phone call at that moment.

Person on phone: Doctor, I have a terrible toothache. Can I visit your office in few minutes?

Dentist : Sorry sir! I am booked up. I have to fill up 18 cavities this afternoon.

Cook!

Jaya was little upset about his husband as he was not helping in cooking. She felt that she was working too hard for party preparations.

Jaya: (screamed) Vivek, Anyone would think that you have employed me as a cook for you in this house?

Vivek: (softly responded) Don't worry! They wouldn't. Especially after they have eaten the food.

Jaya: ?!\$#☹

Horseshoe!

First person : I found a horseshoe – that means good luck!

Seond person: It may mean good luck to you, but some poor horse must be running around in its socks!

First person : ☹

Poor person!

Meena : What would you do if you find a million dollars?

Leena : If it is a poor person who lost the money, I will return it.

Meena : ☹

Swimming - Socks!

Visitor : Do you always swim with your socks on?

Learner : No! Only when the water is cold.

Visitor: ☹

SAFETY-FIRST

Police : Madam! Are you doing OK? It is a bright day and there is no much traffic also. How come you hit that person?

Driver : I was looking at the 'safety-first' instruction and didn't notice that guy.

Police : ☹

Let us see some more jokes in next issue. Till then, Good Bye!

Salutation to your solution!

- Sudhakar Attili.

1. Anand, Vivek and Swaroop participated in a running race. The distance was one and a half kilometer (1500 meters). Anand beat Vivek by 30 meters and Swaroop by 100 meters. Can you find by how much Vivek beat Swaroop, assuming that they both ran as before?
2. Study the logical pattern OTTFFSSEN. What are the next nine letters? Explain.
3. An automatic cutter machine takes one second to cut a one foot ribbon out of a 100 feet bundle. How long will it take to cut the whole bundle into 100 pieces?
4. If $25 \times P = 25 + P$, What is the value of P?
5. Can you find the recurring decimal places of fraction $1/27$ and $1/37$? What is special about it?
6. If $2^n = n^2$, what is the value of n other than 2?

You can send your answers either by post or by e-mail to the below address on or before September 15th.

Mailing Address:

Editor, Sanghamitra,

e-mail :

Can you figure out?

How many quarters can you place touching around quarter?

How many dimes can you place touching another dime?

See the answer with a magnifying glass here → [Click Here](#)

In the below boxes fill in numbers as guided .

1. Fill in the boxes one and two with numbers 1 to 9, 5 to 13 respectively.
2. Sum of numbers in each row, column and diagonals should be same.
3. Fill the third box by adding the numbers from box one and box two.
4. Does the third box satisfy the condition “the sum of numbers in each row, column and diagonal should be same”

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Kids' Corner

- Sudhakar Attili.

Message from a teacher

Once a teacher took his students to a forest. He wanted to explain to his students about values. While going, he noticed some small plants, some moderately small plants, reasonably big plants and lots of big trees. He stopped for a while and all of his students surrounded him.

The teacher planned something in his mind and asked some of his students to pluck the small plants. They pulled them with very little effort. Then he turned to another set of students and asked them to pluck the moderately small plants. They also plucked the plants with little effort. Now he called another batch to pluck big plants. With a little difficulty they plucked those plants. Then he turned to the rest of the students and asked them to pluck the big tree. They tried for a while, but could not even shake the tree.



Then the teacher explained about habits. If a person possesses a bad habit, he should get rid of it immediately. Otherwise, it will become stronger and stronger; getting rid of it becomes harder at that point. He reminded the students a proverb which says "If you can't bend a plant, how can you bend it when it becomes a tree?"

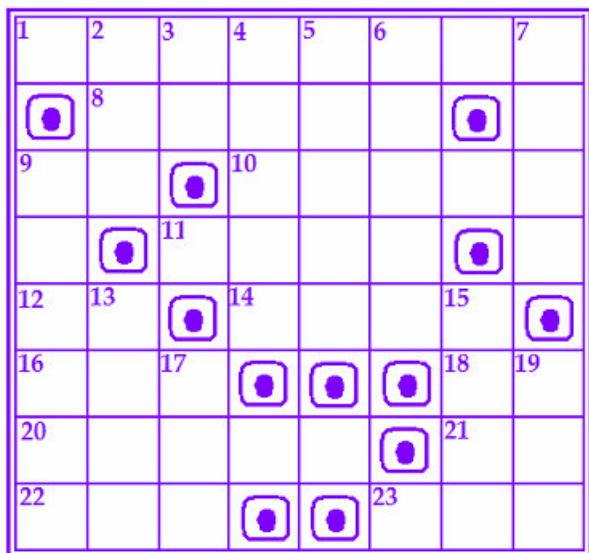
What do we learn from this story?

We need to get rid of bad habits in the initial stage itself; Once they gain strength, it is hard to get rid of them.

Can you find the word?

1. This is a 7 character word and is an item often used in birthday parties.
2. First 4 characters form a spherical object useful for playing.
3. Two different alphabets come in pairs.
4. First, Fifth, Sixth and Seventh characters form a word which means 'granting a wish'.
5. Second, Third and Fourth form a word which means 'You, Me and everyone'.

Cross Word for kids



Across :

1. A person unfamiliar. (8)
8. One who gives or donates something. (5)
9. Abbreviation of 'Internet Protocol'. (2)
10. Seize and hold; Understand. (5)
11. Networks of fine strands made by a spider etc. Plural of last 'w' in www. (4)
12. State abbreviation of Virginia. (2)
14. Possibility of meeting danger or suffering harm. (4)
16. An organ in human body, a palindrome too. (3)
18. State abbreviation of Illinois. (2)
20. Weapon with a long blade and a hilt. (5)

21. **From** is to sender, what is to receiver. (2)
22. White or yellow substance found in animal bodies and certain seeds, right to left. (3)
23. Wooden or metal pin; Clip for holding clothes. (3)

Down :

2. Significant letters of 'pendant'; Starting of 'Pen', 'd', ending of 'Ant'. Upside down. (3)
3. A conjunction, as an alternative. Bottom to top. (2)
4. Extreme displeasure, seen in 'Hanger'. (5)
5. Our body gets instructions from this organ. Characters jumbled. (5)
6. What do we find in Savannah lands? (5)
7. Strong thick cord. (4)
9. Use money to buy shares or property etc. to earn interest or gain profit. (6)
13. Keeping at a distance. Realizing divinity in everyone is 'a way' to live blissfully. Characters jumbled. (4)
15. It has a tail, but not a monkey. It flies, but not a bird. It flies relying on string. (4)
17. Abbreviation of 'End Of File'. (3)
19. Piece cut from a trunk or branch of a tree. (3)

Send your answers (Crossword and 'Can you find the word?') to the below address or e-mail on or before *September 15th*.

e-mail : susri@sanghamitra.org Ph. (314) 395-9516

Mailing Address:

Editor, Sanghamitra, 12176 Vivacite Walk, Apt# B, Saint Louis, MO 63146.

Question Gallery

- Sreenivasa Rao Ainapurapu.

1. In which country is the river '**Pearl or Guangdong**' flowing?
2. Who wrote '**Savitri**'?
3. Where is '**Canary Wall**' located?
4. What is the significance of '**Folic acid**' in human body?
5. What is the capital of the country '**Morocco**'?
6. What is '**Cryptology**'?
7. How many grams is equal to one '**Ounce**'?
8. What is the symbol for '**Selenium**'?
9. In which area '**Nikola Tesla**' is know for?
10. In which subject do we use the word '**Phobia**'?

You can send your answers either by postal mail or e-mail to the below address on or before September 15th.

Mailing address :

Editor, Sanghamitra,

E-Mail address:

or

Mountain Climbers

Two mountain climbers finally reached the top of mountain after their exhausting journey.

The first one said "Even though it is hard and life threatening, it's worthy. We can plant our national flag here. Give me the flag, please!"

Second one stared at him in surprise and said "I thought you brought it!"

Carrying two flags might have been better rather than having none.

Solutions & Winners

Kids' 'X' Word Solution

1 R	2 O	3 S	4 E	◎	5 P
6 I	N	S	E	7 C	T
8 V	E	R	S	E	S
E	◎	◎	◎	9 N	O
10 R	11 E	12 S	13 E	T	◎
◎	14 I	N	G	◎	15 O

Kids Challenge!

Did you find the word?
It is "TRUST".

Smart kids :

1. Sravya Ainapurapu, St. Louis.
2. Sruthi Ainapurapu, St. Louis.
3. D. Aditya, Lucknow.
4. P. Sneha, Delhi.
5. B. Sagar, Nagpur.

Kids' 'X'word Winners:

1. Sruthi Ainapurapu, St. Louis.
2. Sravya Ainapurapu, St. Louis.
3. D. Aditya, Lucknow.

Salutation to your solution! - Answers

1. The wholesale dealer buys hundred-20lb coffee bags at the rate of \$5/lb. So the total cost is $100 \times 20 \times 5 = \10000 . The wholesale dealer sells the bags at \$6.25/lb, \$6.35/lb, \$6.4/lb respectively if they buy minimum of 50-20 pound bags, 30-20pound bags and 20-20 pound bags. The retail shop owners bought 50, 30, 20 bags each weighing 20 pounds. So those retail shop owners' cost price of coffee powder is \$6.25, \$6.35, \$6.40 respectively. Adding profit to the cost price, the retailers' advertised prices are \$6.56/lb, \$6.86 and \$7.04/lb respectively. As per the promotion, when customer buys 5 pounds, he gets another pound free. So the customer needs to buy 7 pounds. First retail shop owner is selling coffee at the rate of \$6.56/lb, which is cheaper. So 8 pounds cost approximately \$45.94.
2. 4999950000.

Solution: Let us take sum of numbers from 1 to 9, which is equal to 45.

(The formula for sum of n natural numbers = $n(n+1)/2$.)

Sum of numbers from 1 to 99, which is equal to 4950.

Sum of numbers from 1 to 999, which is equal to 499500.

Thus, sum of numbers from 1 to 99999 is 4999950000.

3. (a) 72.

(b) 294.

Solution:

- (a) 2, 6, 12, 20, 30, 42, 56 follows the equation $(n^2 + n)$. In this sequence, the next number is $(8^2 + 8)$, which is equal to 72.

- (b) The formula used here $n^2(n-1)$. Following this pattern ...

$$1^2 \times 0 = 0; \quad 2^2 \times 1 = 4; \quad 3^2 \times 2 = 18; \quad 4^2 \times 3 = 48$$

$$5^2 \times 4 = 100; \quad 6^2 \times 5 = 180 \quad 7 \times 6 = 294.$$

4. 6006 square cm.

Solution: For finding the area of metal plate, we need to find the total surface area of the capsule. The total surface area of capsule is equal to the lateral surface area of cylindrical portion and surface areas of hemispheres. $r = 21/2$; $h = 70$

$$\text{Lateral surface area of cylinder} = 2 \times \pi \times r \times h = 2 \times (22/7) \times (21/2) \times 70 \\ = 4620 \text{ Sq. Cm.}$$

Adding the lateral surface area of two hemispheres (not considering the base) is equal to the lateral surface area of sphere.

$$\text{Lateral surface area of sphere} = 4 \times \pi \times r^2 = 4 \times (22/7) \times (21/2)^2 \\ = 1386 \text{ Sq. Cm.}$$

So area of metal plate required = 6006 Sq. Cm.

5. Six circles.

Solution:

Whatever may be the diameter of the circle, when we draw circles of same diameter around it touching the base circle and adjacent circles, only six circles will be accommodated. Logically thinking, The angle at the center is 360° . When we divide the angle into 6 equal parts, then each of the circle (touching the original circle) will fit into one part. You can try this with quarters, dimes etc.

Salutation to your solution! - Winners:

1. Vaasanti Maghapu, California.
2. Ravi Bhagavatula, Chicago.
3. T. Anoop, East Godavari, India.

Question Gallery! - We ourselves disclosing answers!!

1. Thames river flows through London, England.
2. The author of 'My music, my love' is Ravi Shankar.
3. Kandla is a seaport in Kutch District of Gujarat state in western India. Located on the Gulf of Kutch, it is one of major ports on west coast.
4. In Roman number notation, **XCIX** is equal to 99.
5. The capital of Barbados is Bridgetown.
6. Pomology deals with raising fruit trees and productivity of fruits.
7. One Terabyte is equal to 10^{12} bytes. 1 TeraByte = 10^{12} Bytes.
8. Symbol of Curium is Cm.
9. Sir Alexander Fleming (6 August 1881 – 11 March 1955) was a Scottish biologist and pharmacologist. Fleming published many articles on bacteriology, immunology and chemotherapy. His best-known achievements are the discovery of the enzyme lysozyme in 1923 and the antibiotic substance penicillin from the fungus *Penicillium notatum* in 1928, for which he shared the Nobel Prize in Physiology or Medicine in 1945 with Howard Walter Florey and Ernst Boris Chain.
10. Ampere is used in Physics. Ampere is the unit used to measure electric current.

Question Gallery! – Winners : 1. Vaasanti Maghapu, California.

2. Sreekanth Gutala, St. Louis.
3. Ravi Bhagavatula, Chicago.
4. T. Anoop, East Godavari Dt.

Concluding word

Dear readers,

Sanghamitra wishes all readers for a wonderful new academic year. This year Sanghamitra is targeting more universities, middle and high schools to introduce the concepts of Vedic Mathematics. For the first time it organized classes in bay area, California.

We request you to send articles either a science article, news article, about culture or anything that you think is useful for the readers. Send any articles or your feedback to .

We are working on giving prizes to winners by contacting friends' network. Send your answers before September 15th either by E-mail or Postal mail. Encourage your kids to do Kids' Xword and other challenges. Tell your friends about Sanghamitra.

Note : Answers should be sent on or before **September 15th**.

Sanghamitra Team.

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Saint Louis,

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or